

# Aman Gairola

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Portfolio | LinkedIn | GitHub | LeetCode

## EDUCATION

### Bachelor of Technology in Computer Science and Engineering

Shivalik College of Engineering | Dehradun, India

Expected: May 2026

CGPA: 7.0/10

### Diploma in Information Technology

Government Polytechnic Gauchar | Uttarakhand, India

June 2020 – May 2023

CGPA: 9.0/10 (Distinction)

## TECHNICAL SKILLS

**Programming Languages:** Java, JavaScript (ES6+), TypeScript, C++, SQL, HTML5, CSS3

**Frontend Development:** React.js, Redux, Tailwind CSS, Material-UI, Responsive Web Design

**Backend Development:** Node.js, Express.js, RESTful APIs, WebSockets, Socket.io, JWT Authentication

**Database Management:** MongoDB, Firebase (Firestore, Authentication), MySQL, PostgreSQL

**Developer Tools:** Git, GitHub, Docker, Postman, VS Code, Vercel, Render, CI/CD Pipelines

**Core Competencies:** Data Structures & Algorithms, Object-Oriented Programming, System Design, Full Stack Development, Problem Solving, Performance Optimization, Agile Methodologies, Test-Driven Development

## PROFESSIONAL EXPERIENCE

### Software Engineering Intern

June 2023 – August 2023

VerveGen Tech Pvt Ltd

Dehradun, India

- Engineered 15+ modular and reusable React.js components utilizing JavaScript ES6+ and Tailwind CSS, enhancing code reusability by 40% and accelerating feature delivery by 2 weeks across multiple product sprints
- Optimized front-end performance through strategic code refactoring, lazy loading, and bundle size reduction, achieving 25% faster initial page load times and improving Lighthouse performance score by 30 points

## TECHNICAL PROJECTS

### Tracker.io - Gamified DSA Progress Tracker | MERN Stack, OAuth 2.0, JWT

GitHub | Live

- Built full-stack productivity platform with Product-Led Growth architecture featuring frictionless Guest Mode for instant dashboard access and Google OAuth 2.0 integration for seamless one-click authentication
- Engineered hybrid JWT authentication with automatic Access/Refresh token rotation using custom Axios interceptors and HTTP-Only secure cookies, eliminating unexpected logouts and ensuring persistent user sessions
- Implemented GitHub-style activity heatmap with custom streak calculation algorithm and React Context API for dynamic state management, enabling real-time UI transitions between Guest and authenticated modes without page reloads

### Collabb - Real-Time Collaborative Code Editor | React.js, Node.js, Socket.io

GitHub | Live

- Developed full-stack collaborative coding platform supporting 50+ concurrent users with real-time code synchronization using WebSocket technology, achieving sub-200ms latency for seamless multi-user editing experiences
- Architected scalable backend infrastructure using Node.js and Express.js with conflict resolution algorithms, ensuring 99.9% uptime through automated health monitoring and load balancing strategies
- Implemented user session management and room-based architecture with Socket.io, enabling secure workspace creation and dynamic participant tracking across distributed coding sessions

### Folyo - Full-Stack Blog Platform | MERN Stack, JWT, GitHub Actions

GitHub | Live

- Architected scalable MERN application with 20+ RESTful API endpoints handling CRUD operations for posts, comments, and user profiles with comprehensive input validation and error handling middleware
- Implemented role-based access control (RBAC) with JWT authentication and authorization, securing admin dashboards and enabling granular permission management for content moderation
- Established automated CI/CD pipeline using GitHub Actions for continuous integration and deployment, reducing deployment time by 83% from 30 minutes to under 5 minutes with zero-downtime releases

## ACHIEVEMENTS & CERTIFICATIONS

**Competitive Programming:** Solved 300+ Data Structures and Algorithms problems on LeetCode with focus on optimal time-space complexity analysis and algorithm design patterns

**Open Source Contribution:** Active contributor on GeeksforGeeks with 50+ accepted solutions in Java and JavaScript, demonstrating strong problem-solving capabilities

**Professional Certification:** Completed "Data Structures and Algorithms in Java" specialization from Coursera with hands-on implementation of core algorithms